Formulations of packing problems

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Learning objectives

At the end of this chapter you will be able to:

- write the ILP model of the Bin Packing Problem and explain the link between the assignment variables and the activation variables;
- write the two LP relaxations and their duals, and prove that their optimal value is $\sum_j w_j/c$;
- round the LP bound up to obtain a lower bound on the number of bins;
- prove that the LP bound is at least one half of the optimal value, and that the factor is tight;
- write the ILP model of the Cutting Stock Problem, with integer variables for the copies of the items.

Prerequisites

- From the module *Linear Programming*, the chapter «Duality theory»: the dual of an LP problem, weak and strong duality.
- From the module *Integer linear programming*, the chapters «Integer linear programming» and «Primal and dual bounds»: ILP problems, LP relaxations and bounds.
- From this module, the chapter «Packing problems»: the problem and its assumptions.

Overview of the family

Introduction: Overview of the family

- In this chapter we write the ILP models of the packing problems: the Bin Packing Problem (BPP) and the Cutting Stock Problem (CSP)
- We first see what the models of the family have in common: the same variables, the same constraints, the same objective function
- The models are the starting point for the LP relaxations, and the relaxations give lower bounds on the number of bins

Bins, assignment variables and activation variables

- The ILP models of the packing problems use an upper bound m on the number of bins and the set $\mathscr{B} = \{1, 2, \ldots, m\}$ of the bins that may be used.
- The **assignment variables** x_{jb} place the items $j \in \mathscr{I}$ into the bins $b \in \mathscr{B}$, and the binary **activation variables** y_b say whether bin b is used.
- Each bin gives one capacity constraint, which also links the two kinds of variables, and the objective function minimizes the number $\sum_{b \in \mathscr{B}} y_b$ of bins used.

Packing models are assignment models

The packing models are **assignment models**: every item, or every copy of an item, is assigned to a used bin, and the items assigned to a bin respect its capacity.

The members of the family change the domain of the assignment variables.

The members of the family

- The table lists the members of the family, with the name of their ILP model and the section that writes it.

Problem	**Model**	**Assignment variables**	**Section**
BPP	$\mathrm{ILP}_{\mathrm{BPP}}$	binary x_{jb}	An ILP model for the Bin Packing Problem
CSP	$\mathrm{ILP}_{\mathrm{CSP}}$	integer x_{jb}, the copies of item j in bin b	An ILP model for the Cutting Stock Problem

Summary

The common structure:

- an upper bound m on the number of bins, and the set $\mathscr{B}$ of the bins
- assignment variables x_{jb} and binary activation variables y_b
- one capacity constraint per bin, and the number of used bins as objective function

The members:

- BPP: binary assignment variables
- CSP: integer assignment variables, which count the copies

An ILP model for the Bin Packing Problem

Introduction: An ILP model for the Bin Packing Problem

- The problem of this section is the Bin Packing Problem (BPP), defined in the chapter «Packing problems»
- We introduce the variables, write the $\mathrm{ILP}_{\mathrm{BPP}}$ model and describe its constraints
- We explain how the capacity constraints link the two kinds of variables, and we write the model of a small instance

The assignment variables

- Given an upper bound $m \in \mathbb{Z}_{\geq 2}$ on the number of bins used in any optimal BPP solution, we introduce the following mn **binary variables**:

$$x_{jb} = \begin{cases} 1 & \text{if item } j \text{ is packed in bin } b \\ 0 & \text{otherwise} \end{cases}$$

$$\forall j \in \underbrace{\{1, 2, \ldots, n\}}_{=\mathscr{J}}, \quad \forall b \in \underbrace{\{1, 2, \ldots, m\}}_{=\mathscr{B}},$$

The activation variables

- We also introduce the following m **binary variables**:

$$y_b = \begin{cases} 1 & \text{if bin } b \text{ is used} \\ 0 & \text{otherwise} \end{cases} \qquad \forall b \in \underbrace{\{1, 2, \ldots, m\}}_{=\mathcal{B}}.$$

- The number n of items is a valid upper bound on the minimum number of bins needed to pack all the items since, in the worst possible case, each item is packed in a different bin.
- Also the number of bins used in any feasible BPP solution is a valid upper bound.

The $\mathrm{ILP}_{\mathrm{BPP}}$ model

Using these variables, an Integer Linear Programming (ILP) model for the Bin Packing Problem (BPP) reads as follows:

$$\min \quad \sum_{b \in \mathscr{B}} y_b$$

$$\text{subject to} \quad \sum_{b \in \mathscr{B}} x_{jb} = 1, \qquad \forall j \in \mathscr{I},$$

$$-\sum_{j \in \mathscr{I}} w_j \, x_{jb} + c \, y_b \geq 0, \qquad \forall b \in \mathscr{B},$$

$$x_{jb} \in \{0,1\}, \quad \forall j \in \mathscr{I}, \; \forall b \in \mathscr{B},$$

$$y_b \in \{0,1\}, \qquad \forall b \in \mathscr{B}.$$

This ILP model is referred to as the $\mathrm{ILP}_{\mathrm{BPP}}$ model.

Description of the objective function and constraints

- The linear objective function minimizes the total number of used bins.
- The constraints $\sum_{b \in \mathcal{B}} x_{jb} = 1$ ensure that each item is packed in exactly one bin, so that all items are packed (n linear constraints).
- The capacity constraints link the assignment variables with the activation variables and impose the capacity restrictions: if bin b is used, the weight of its items is at most c; if it is not used, it contains no item (m linear constraints).
- Finally, the constraints $x_{jb} \in \{0, 1\}$ and $y_b \in \{0, 1\}$ define the variables of the ILP model.

The capacity constraints do two jobs at once:
they limit the weight in each bin and they link x_{jb} to y_b.

Let us see the link between the variables in detail.

Explanation of the link between the variables (1/2)

- **If at least one item is packed in bin b, then the bin is used.** The logical implication is

$$(x_{1b} \text{ OR } x_{2b} \text{ OR } \ldots \text{ OR } x_{nb}) \quad \Longrightarrow \quad y_b.$$

  Since $w_j > 0$, packing an item in bin b gives $\sum_{j \in \mathscr{J}} w_j x_{jb} > 0$, and the constraint $\sum_{j \in \mathscr{J}} w_j x_{jb} \leq c\, y_b$ forces $y_b = 1$.

- **If bin b is not used, then it is empty.** This is the contrapositive of the previous implication. For $y_b = 0$ the constraint reads $\sum_{j \in \mathscr{J}} w_j x_{jb} \leq 0$, and since $w_j > 0$ we obtain $x_{jb} = 0$ for every item j.

Explanation of the link between the variables (2/2)

- **If bin b is used, then it contains at least one item, but only in an optimal solution.**
- The constraints do not impose it.
- However, if $x_{jb} = 0$ for all j and $y_b = 1$, setting $y_b = 0$ keeps all the constraints satisfied and decreases the objective value by one.
- Hence, in every optimal solution, empty bins are not counted as used.

Now we write the $\mathrm{ILP}_{\mathrm{BPP}}$ model of a small instance, with three items, and we look at its optimal solutions.

Example (1/4): the $\mathrm{ILP}_{\mathrm{BPP}}$ model

- We consider a bin of capacity $c = 10$ and a set $\mathscr{I} = \{1, 2, 3\}$ of 3 items ($n = 3$), with the following item weights:

	$j = 1$	$j = 2$	$j = 3$
w_j	7	6	3

- In the rest of the chapter we refer to this BPP instance as the **example instance**.
- We write the $\mathrm{ILP}_{\mathrm{BPP}}$ model with $m = 3$.

Example (2/4): the $\mathrm{ILP}_{\mathrm{BPP}}$ model

The $\mathrm{ILP}_{\mathrm{BPP}}$ model with $m = 3$ for this BPP instance is:

$$
\begin{array}{lrrrrcl}
\text{min} & y_1 & + y_2 & + y_3 & & & \\
\text{subject to} & x_{11} & + x_{12} & + x_{13} & & = & 1 \\
 & x_{21} & + x_{22} & + x_{23} & & = & 1 \\
 & x_{31} & + x_{32} & + x_{33} & & = & 1 \\
 & -7x_{11} & - 6x_{21} & - 3x_{31} & + 10y_1 & \geq & 0 \\
 & -7x_{12} & - 6x_{22} & - 3x_{32} & + 10y_2 & \geq & 0 \\
 & -7x_{13} & - 6x_{23} & - 3x_{33} & + 10y_3 & \geq & 0 \\
 & x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}, x_{31}, x_{32}, x_{33} & & & & \in & \{0, 1\} \\
 & y_1, y_2, y_3 & & & & \in & \{0, 1\}
\end{array}
$$

Example (3/4): the $\mathrm{ILP}_{\mathrm{BPP}}$ model

An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{ILP}_{\mathrm{BPP}})$ is:

$$\tilde{x}_{11} = \tilde{x}_{31} = 1, \quad \tilde{x}_{12} = \tilde{x}_{13} = \tilde{x}_{21} = 0, \quad \tilde{x}_{22} = 1,$$

$$\tilde{x}_{23} = \tilde{x}_{32} = \tilde{x}_{33} = 0, \quad \tilde{y}_1 = \tilde{y}_2 = 1, \quad \tilde{y}_3 = 0$$

Another optimal solution, obtained by permuting the indices of bins 1 and 2, is:

$$\tilde{x}_{12} = \tilde{x}_{32} = 1, \quad \tilde{x}_{11} = \tilde{x}_{13} = \tilde{x}_{22} = 0, \quad \tilde{x}_{21} = 1,$$

$$\tilde{x}_{23} = \tilde{x}_{31} = \tilde{x}_{33} = 0, \quad \tilde{y}_1 = \tilde{y}_2 = 1, \quad \tilde{y}_3 = 0.$$

Example (4/4): the $\mathrm{ILP}_{\mathrm{BPP}}$ model

The optimal value is:

$$z(\mathrm{ILP}_{\mathrm{BPP}}) = 2$$

The associated optimal solution of the BPP is given by the packing patterns

$$\tilde{P}_1 = \{1, 3\} \quad \text{and} \quad \tilde{P}_2 = \{2\}$$

The optimal solution is not unique: also the packing patterns $\{1\}$ and $\{2, 3\}$ use two bins.

Summary (1/2)

The $\mathrm{ILP}_{\mathrm{BPP}}$ model:

- mn binary assignment variables x_{jb} and m binary activation variables y_b
- each item in exactly one bin, one capacity constraint per bin
- minimize the number of used bins

Summary (2/2)

The link between the variables:

- a non-empty bin is used, an unused bin is empty
- an empty bin is not counted as used, but only in an optimal solution

The example instance: $z(\mathrm{ILP}_{\mathrm{BPP}}) = 2$, with patterns $\{1, 3\}$ and $\{2\}$

A first Linear Programming relaxation

Introduction: A first Linear Programming relaxation

- We relax the binary domains of the $\mathrm{ILP}_{\mathrm{BPP}}$ model and obtain an LP model, the $\mathrm{LP\text{-}0}_{\mathrm{BPP}}$ model
- Its optimal value is a lower bound on the number of bins: we compute it on the example instance and then for every instance
- We write the dual model and prove that the optimal value is $\sum_j w_j/c$, with two different proofs

The LP-0_{BPP} model

To obtain a Linear Programming (LP) relaxation of the $\mathrm{ILP}_{\mathrm{BPP}}$ model, we relax the constraints $x_{jb} \in \{0, 1\}$ and $y_b \in \{0, 1\}$ with the following ones:

$$x_{jb} \geq 0, \quad \forall j \in \mathscr{I}, \ \forall b \in \mathscr{B},$$

$$y_b \geq 0, \qquad \forall b \in \mathscr{B}.$$

This LP model is referred to as the LP-0_{BPP} model.

Example (1/2): the LP-0_{BPP} model

The LP-0_{BPP} model for the example instance is:

$$
\begin{array}{lrrrrrcl}
\text{min} & y_1 & +y_2 & +y_3 & & & & \\
\text{subject to} & x_{11} & +x_{12} & +x_{13} & & & = & 1 \\
 & x_{21} & +x_{22} & +x_{23} & & & = & 1 \\
 & x_{31} & +x_{32} & +x_{33} & & & = & 1 \\
 & -7x_{11} & -6x_{21} & -3x_{31} & & +10y_1 & \geq & 0 \\
 & -7x_{12} & -6x_{22} & -3x_{32} & & +10y_2 & \geq & 0 \\
 & -7x_{13} & -6x_{23} & -3x_{33} & & +10y_3 & \geq & 0 \\
 & x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}, x_{31}, x_{32}, x_{33} & & & & & \geq & 0 \\
 & y_1, y_2, y_3 & & & & & \geq & 0
\end{array}
$$

Example (2/2): the LP-0_{BPP} model

An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP\text{-}0_{BPP}})$ is:

$$\tilde{x}_{11} = \tilde{x}_{12} = \tilde{x}_{13} = \tilde{x}_{21} = \tilde{x}_{22} = \tilde{x}_{23} = \tilde{x}_{31} = \tilde{x}_{32} = \tilde{x}_{33} = \frac{1}{3},$$

$$\tilde{y}_1 = \tilde{y}_2 = \tilde{y}_3 = \frac{8}{15}$$

The optimal value $z(\mathrm{LP\text{-}0_{BPP}})$ is $\frac{8}{5}$. Accordingly, we have the lower bound:

$$\frac{8}{5} \leq z(\mathrm{ILP_{BPP}})$$

and, since $z(\mathrm{ILP_{BPP}})$ is an integer, $z(\mathrm{ILP_{BPP}}) \geq \lceil \frac{8}{5} \rceil = 2$.

The dual variables of the $\mathrm{LP\text{-}0_{BPP}}$ model

Using

- a free dual variable μ_j for each constraint $\sum_{b \in \mathscr{B}} x_{jb} = 1$ $(j \in \mathscr{I})$
- a non-negative dual variable π_b for each capacity constraint $(b \in \mathscr{B})$

we write the dual model of the $\mathrm{LP\text{-}0_{BPP}}$ model.

The dual model $D(\text{LP-0}_{\text{BPP}})$

The dual model of the LP-0_{BPP} model reads as follows:

$$\max \quad \sum_{j \in \mathscr{I}} \mu_j$$

$$\begin{aligned}
\text{subject to} \quad \mu_j - w_j \, \pi_b &\leq 0, \quad \forall j \in \mathscr{I}, \ \forall b \in \mathscr{B}, \\
c \, \pi_b &\leq 1, \qquad \forall b \in \mathscr{B}, \\
\mu_j &\gtreqless 0, \qquad \forall j \in \mathscr{I}, \\
\pi_b &\geq 0, \qquad \forall b \in \mathscr{B}.
\end{aligned}$$

This LP model is referred to as the dual $D(\text{LP-0}_{\text{BPP}})$ model.

Example (1/3): the $D(\text{LP-0}_{\text{BPP}})$ model

The dual $D(\text{LP-0}_{\text{BPP}})$ model for the example instance is (objective and first constraints):

$$
\begin{array}{rrrrrrrrrrrr}
\max & \mu_1 & +\mu_2 & +\mu_3 & & & & & & & \\
\text{subject to} & \mu_1 & & & -7\pi_1 & & & & & \leq & 0 \\
 & \mu_1 & & & & -7\pi_2 & & & & \leq & 0 \\
 & \mu_1 & & & & & -7\pi_3 & & & \leq & 0 \\
 & & \mu_2 & & -6\pi_1 & & & & & \leq & 0 \\
 & & \mu_2 & & & -6\pi_2 & & & & \leq & 0 \\
 & & \mu_2 & & & & -6\pi_3 & & & \leq & 0 \\
 & & & \mu_3 & -3\pi_1 & & & & & \leq & 0 \\
 & & & \mu_3 & & -3\pi_2 & & & & \leq & 0 \\
 & & & \mu_3 & & & -3\pi_3 & & & \leq & 0
\end{array}
$$

Example (2/3): the $D(\text{LP-0}_{\text{BPP}})$ model

The model continues with the remaining constraints and the domains:

$$
\begin{array}{cccccccccc}
 & & & 10\pi_1 & & & & \leq & 1 \\
 & & & & 10\pi_2 & & & \leq & 1 \\
 & & & & & & 10\pi_3 & \leq & 1 \\
\mu_1, & \mu_2, & \mu_3 & & & & & \geq & 0 \\
 & & & \pi_1, & & \pi_2, & & \geq & 0
\end{array}
$$

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Example (3/3): the $D(\text{LP-0}_{\text{BPP}})$ model

An optimal solution $(\tilde{\boldsymbol{\mu}}, \tilde{\boldsymbol{\pi}})$ of $D(\text{LP-0}_{\text{BPP}})$ is:

$$\tilde{\mu}_1 = \frac{7}{10}, \quad \tilde{\mu}_2 = \frac{3}{5}, \quad \tilde{\mu}_3 = \frac{3}{10}, \quad \tilde{\pi}_1 = \tilde{\pi}_2 = \tilde{\pi}_3 = \frac{1}{10}$$

The optimal value $z(D(\text{LP-0}_{\text{BPP}}))$ is $\frac{8}{5}$.

On the example instance the optimal value is $\frac{8}{5} = \frac{7+6+3}{10}$.

We now give optimal primal and dual solutions for **every** BPP instance.

Optimal solutions of LP-0_{BPP} and of its dual

Proposition 1

An optimal solution of the primal LP-0_{BPP} *model is:*

$$\tilde{x}_{jb} = \frac{1}{m}, \qquad \forall j \in \mathscr{I},\ \forall b \in \mathscr{B},$$

$$\tilde{y}_b = \frac{\sum_{j \in \mathscr{I}} w_j}{m\, c}, \qquad \forall b \in \mathscr{B},$$

and an optimal solution of the dual $D(\text{LP-}0_{\mathrm{BPP}})$ *model is:*

$$\tilde{\mu}_j = \frac{w_j}{c}, \quad \forall j \in \mathscr{I},$$

$$\tilde{\pi}_b = \frac{1}{c}, \quad \forall b \in \mathscr{B}.$$

The optimal value of LP-0_{BPP}

Corollary 1

The optimal value of the primal LP-0_{BPP} *and dual* $D(\text{LP-0}_{\text{BPP}})$ *models is:*

$$z(\text{LP-0}_{\text{BPP}}) = \frac{\sum_{j \in \mathscr{J}} w_j}{c} = z(D(\text{LP-0}_{\text{BPP}}))$$

How the proof proceeds

- The proof proceeds in three steps:
  1. We show that the primal solution $\tilde{x}_{jb}$, $\tilde{y}_b$ of the proposition is a feasible solution for the primal $\mathrm{LP\text{-}0_{BPP}}$ model;
  2. We show that the dual solution $\tilde{\mu}_j$, $\tilde{\pi}_b$ of the proposition is a feasible solution for the dual $D(\mathrm{LP\text{-}0_{BPP}})$ model;
  3. We show that both feasible solutions have the same objective function value.

- Therefore, by LP Weak Duality, the two solutions are a pair of optimal primal and dual solutions.

Proof (1/4)

1. For every $j \in \mathscr{J}$, we have:

$$\sum_{b \in \mathscr{B}} \tilde{x}_{jb} = \sum_{b \in \mathscr{B}} \frac{1}{m} = \frac{m}{m} = 1$$

thus the constraints $\sum_{b \in \mathscr{B}} x_{jb} = 1$ are satisfied.

For every $b \in \mathscr{B}$, we have:

$$-\sum_{j \in \mathscr{J}} w_j \, \tilde{x}_{jb} + c \, \tilde{y}_b = -\sum_{j \in \mathscr{J}} w_j \, \frac{1}{m} + c \, \frac{\sum_{j \in \mathscr{J}} w_j}{m \, c} = 0$$

thus the capacity constraints are satisfied.

Proof (2/4)

1. We have $\tilde{x}_{jb} \geq 0$ and $\tilde{y}_b \geq 0$, for every $j \in \mathscr{I}$ and every $b \in \mathscr{B}$, thus the non-negativity constraints are satisfied.

   Therefore, $\tilde{x}_{jb}$, $\tilde{y}_b$ provide a feasible solution for the primal $\mathrm{LP\text{-}0_{BPP}}$ model.

2. For every $b \in \mathscr{B}$ and every $j \in \mathscr{I}$, we have:

$$\tilde{\mu}_j - w_j \, \tilde{\pi}_b = \frac{w_j}{c} - \frac{w_j}{c} = 0$$

   thus the constraints $\mu_j - w_j \, \pi_b \leq 0$ are satisfied.

Proof (3/4)

2. For every $b \in \mathscr{B}$, we have:

$$c\ \tilde{\pi}_b = \frac{c}{c} = 1$$

thus the constraints $c\ \pi_b \leq 1$ are satisfied.

We have $\tilde{\mu}_j \geq 0$ and $\tilde{\pi}_b \geq 0$, for every $b \in \mathscr{B}$ and every $j \in \mathscr{I}$, thus the domains of the dual variables are respected.

Therefore, $\tilde{\mu}_j$, $\tilde{\pi}_b$ provide a feasible solution for the dual $D(\text{LP-0}_{\text{BPP}})$ model.

Proof (4/4)

3. For the objective functions, we have:

$$\sum_{b \in \mathscr{B}} \tilde{y}_b = \sum_{b \in \mathscr{B}} \frac{\sum_{j \in \mathscr{I}} w_j}{m\ c} = \frac{\sum_{j \in \mathscr{I}} w_j}{c} = \sum_{j \in \mathscr{I}} \tilde{\mu}_j$$

thus the primal and dual objective function values are equal.

Therefore, by LP Weak Duality, the two solutions are a pair of optimal primal and dual solutions. □

An alternative proof

- This is an alternative proof that does not rely on the dual formulation but instead reasons directly on the minimum objective function value imposed by the constraints.
- The idea: summing the capacity constraints gives a lower bound on $\sum_{b \in \mathcal{B}} y_b$ that holds for every feasible solution; then we find a feasible solution that attains it.

The optimal value of LP-0_{BPP}, without the dual

Proposition 2

An optimal solution of the primal LP-0_{BPP} *model is:*

$$\tilde{x}_{jb} = \frac{1}{m}, \qquad \forall j \in \mathscr{I},\ \forall b \in \mathscr{B},$$

$$\tilde{y}_b = \frac{\sum_{j \in \mathscr{I}} w_j}{m\ c}, \qquad \forall b \in \mathscr{B},$$

and the optimal value is

$$z(\text{LP-}0_{\mathrm{BPP}}) = \frac{\sum_{j \in \mathscr{I}} w_j}{c}.$$

Proof (1/3)

Summing the capacity constraints over all $b \in \mathscr{B}$ gives

$$c \sum_{b \in \mathscr{B}} y_b \;\geq\; \sum_{b \in \mathscr{B}} \sum_{j \in \mathscr{I}} w_j x_{jb} = \sum_{j \in \mathscr{I}} w_j \underbrace{\sum_{b \in \mathscr{B}} x_{jb}}_{=1} = \sum_{j \in \mathscr{I}} w_j,$$

hence

$$\sum_{b \in \mathscr{B}} y_b \;\geq\; \frac{1}{c} \sum_{j \in \mathscr{I}} w_j.$$

Therefore every feasible solution of $\mathrm{LP\text{-}0_{BPP}}$ satisfies this lower bound.

Proof (2/3)

Now consider the point of the proposition. For every $j \in \mathscr{J}$, we have

$$\sum_{b \in \mathscr{B}} \tilde{x}_{jb} = m \, \frac{1}{m} = 1,$$

so the constraints $\sum_{b \in \mathscr{B}} x_{jb} = 1$ hold.

Moreover, for every $b \in \mathscr{B}$, we have

$$\sum_{j \in \mathscr{J}} w_j \tilde{x}_{jb} = \sum_{j \in \mathscr{J}} w_j \, \frac{1}{m} = \frac{1}{m} \sum_{j \in \mathscr{J}} w_j = c \, \tilde{y}_b,$$

so the capacity constraints hold with equality.

Proof (3/3)

Finally, the non-negativity constraints are satisfied by construction. Hence the point of the proposition is a feasible solution for $\mathrm{LP\text{-}0}_{\mathrm{BPP}}$.

The objective value of this point is

$$\sum_{b \in \mathscr{B}} \tilde{y}_b = m \, \frac{\sum_{j \in \mathscr{I}} w_j}{m \, c} = \frac{1}{c} \sum_{j \in \mathscr{I}} w_j,$$

which matches the lower bound above.

Therefore the point is an optimal solution with objective value $\sum_{j \in \mathscr{I}} w_j / c$. □

Summary

The $\mathrm{LP\text{-}0_{BPP}}$ model:

- the binary domains are replaced by $x_{jb} \geq 0$ and $y_b \geq 0$
- on the example instance: $z(\mathrm{LP\text{-}0_{BPP}}) = \frac{8}{5}$, hence $z(\mathrm{ILP_{BPP}}) \geq \lceil \frac{8}{5} \rceil = 2$

The optimal value:

- primal solution $\tilde{x}_{jb} = 1/m$, dual solution $\tilde{\mu}_j = w_j/c$ and $\tilde{\pi}_b = 1/c$
- $z(\mathrm{LP\text{-}0_{BPP}}) = \sum_j w_j/c$, by weak duality or by summing the capacity constraints

A second Linear Programming relaxation

Introduction: A second Linear Programming relaxation

- We relax the binary domains of the $\mathrm{ILP}_{\mathrm{BPP}}$ model keeping the upper bounds $y_b \leq 1$: this is the $\mathrm{LP}_{\mathrm{BPP}}$ model
- We write it on the example instance, write its dual, and show that its optimal value is again $\sum_j w_j/c$
- Then we measure the quality of this bound: it is at least one half of the optimal number of bins, and the factor one half cannot be improved

The $\mathrm{LP}_{\mathrm{BPP}}$ model

To obtain a Linear Programming (LP) relaxation of the $\mathrm{ILP}_{\mathrm{BPP}}$ model, we relax the constraints $x_{jb} \in \{0, 1\}$ and $y_b \in \{0, 1\}$ with the following ones:

$$x_{jb} \geq 0, \quad \forall j \in \mathscr{I}, \forall b \in \mathscr{B},$$

$$y_b \leq 1, \qquad \forall b \in \mathscr{B},$$

$$y_b \geq 0, \qquad \forall b \in \mathscr{B}.$$

This LP model is referred to as the $\mathrm{LP}_{\mathrm{BPP}}$ model.

Example (1/3): the $\mathrm{LP}_{\mathrm{BPP}}$ model

The $\mathrm{LP}_{\mathrm{BPP}}$ model for the example instance is (objective, assignment and capacity constraints):

$$
\begin{array}{rrrrrrr}
\text{min} & y_1 & +y_2 & +y_3 & & & \\
\text{subject to} & x_{11} & +x_{12} & +x_{13} & & = & 1 \\
 & x_{21} & +x_{22} & +x_{23} & & = & 1 \\
 & x_{31} & +x_{32} & +x_{33} & & = & 1 \\
 & -7x_{11} & -6x_{21} & -3x_{31} & +10y_1 & \geq & 0 \\
 & -7x_{12} & -6x_{22} & -3x_{32} & +10y_2 & \geq & 0 \\
 & -7x_{13} & -6x_{23} & -3x_{33} & +10y_3 & \geq & 0
\end{array}
$$

Example (2/3): the $\mathrm{LP}_{\mathrm{BPP}}$ model

The model continues with the upper bounds $y_b \leq 1$ and the non-negativity constraints:

$$
\begin{aligned}
y_1 &\leq 1 \\
y_2 &\leq 1 \\
y_3 &\leq 1 \\
x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}, x_{31}, x_{32}, x_{33} &\geq 0 \\
y_1, y_2, y_3 &\geq 0
\end{aligned}
$$

Example (3/3): the $\mathrm{LP}_{\mathrm{BPP}}$ model

An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_{\mathrm{BPP}})$ is:

$$\tilde{x}_{11} = \tilde{x}_{12} = \tilde{x}_{13} = \tilde{x}_{21} = \tilde{x}_{22} = \tilde{x}_{23} = \tilde{x}_{31} = \tilde{x}_{32} = \tilde{x}_{33} = \frac{1}{3},$$

$$\tilde{y}_1 = \tilde{y}_2 = \tilde{y}_3 = \frac{8}{15}$$

The optimal value $z(\mathrm{LP}_{\mathrm{BPP}})$ is $\frac{8}{5}$.

Accordingly, we have the same lower bound as before.

The dual variables of the $\mathrm{LP}_{\mathrm{BPP}}$ model

Using

- a free dual variable μ_j for each constraint $\sum_{b \in \mathscr{B}} x_{jb} = 1$ $(j \in \mathscr{I})$
- a non-negative dual variable π_b for each capacity constraint $(b \in \mathscr{B})$
- a non-positive dual variable λ_b for each constraint $y_b \leq 1$ $(b \in \mathscr{B})$

we write the dual model of the $\mathrm{LP}_{\mathrm{BPP}}$ model.

The dual model $D(\mathrm{LP}_{\mathrm{BPP}})$

The dual model of the $\mathrm{LP}_{\mathrm{BPP}}$ model reads as follows:

$$\max \quad \sum_{j \in \mathscr{J}} \mu_j + \sum_{b \in \mathscr{B}} \lambda_b$$

$$\begin{aligned}
\text{subject to} \quad & \mu_j - w_j \, \pi_b \leq 0, && \forall j \in \mathscr{J}, \ \forall b \in \mathscr{B}, \\
& c \, \pi_b + \lambda_b \leq 1, && \forall b \in \mathscr{B}, \\
& \mu_j \gtreqless 0, && \forall j \in \mathscr{J}, \\
& \pi_b \geq 0, && \forall b \in \mathscr{B}, \\
& \lambda_b \leq 0, && \forall b \in \mathscr{B}.
\end{aligned}$$

This LP model is referred to as the dual $D(\mathrm{LP}_{\mathrm{BPP}})$ model.

Example (1/3): the $D(\mathrm{LP}_{\mathrm{BPP}})$ model

The $D(\mathrm{LP}_{\mathrm{BPP}})$ model of the example instance is (objective and first constraints):

$$
\begin{array}{rcccccccccc}
\max & \mu_1 & +\mu_2 & +\mu_3 & & & & & +\lambda_1 & +\lambda_2 & +\lambda_3 \\
\text{subject to} & \mu_1 & & & -7\pi_1 & & & & & & \leq 0 \\
& \mu_1 & & & & -7\pi_2 & & & & & \leq 0 \\
& \mu_1 & & & & & -7\pi_3 & & & & \leq 0 \\
& & \mu_2 & & -6\pi_1 & & & & & & \leq 0 \\
& & \mu_2 & & & -6\pi_2 & & & & & \leq 0 \\
& & \mu_2 & & & & -6\pi_3 & & & & \leq 0 \\
& & & \mu_3 & -3\pi_1 & & & & & & \leq 0 \\
& & & \mu_3 & & -3\pi_2 & & & & & \leq 0 \\
& & & \mu_3 & & & -3\pi_3 & & & & \leq 0
\end{array}
$$

Example (2/3): the $D(\mathrm{LP}_{\mathrm{BPP}})$ model

The model continues with the remaining constraints and the domains:

$$
\begin{array}{rrrrrrrrrl}
 & 10\pi_1 & & & & + \lambda_1 & & & & \leq 1 \\
 & & 10\pi_2 & & & & + \lambda_2 & & & \leq 1 \\
 & & & 10\pi_3 & & & & + \lambda_3 & & \leq 1 \\
\mu_1, & \mu_2, & \mu_3 & & & & & & & \gtreqless 0 \\
 & & & \pi_1, & \pi_2, & \pi_3 & & & & \geq 0 \\
 & & & & & & \lambda_1, & \lambda_2, & \lambda_3 & \leq 0
\end{array}
$$

Example (3/3): the $D(\mathrm{LP}_{\mathrm{BPP}})$ model

The optimal value $z(D(\mathrm{LP}_{\mathrm{BPP}}))$ is $\frac{8}{5}$.

An optimal solution $(\tilde{\boldsymbol{\mu}}, \tilde{\boldsymbol{\pi}}, \tilde{\boldsymbol{\lambda}})$ of $D(\mathrm{LP}_{\mathrm{BPP}})$ is:

$$\tilde{\mu}_1 = \frac{7}{10}, \quad \tilde{\mu}_2 = \frac{3}{5}, \quad \tilde{\mu}_3 = \frac{3}{10},$$

$$\tilde{\pi}_1 = \tilde{\pi}_2 = \tilde{\pi}_3 = \frac{1}{10}, \quad \tilde{\lambda}_1 = \tilde{\lambda}_2 = \tilde{\lambda}_3 = 0$$

Valid constraints implied by the model

- The constraints:

$$x_{jb} \leq 1, \qquad \forall j \in \mathscr{J},\ \forall b \in \mathscr{B},$$

are also valid since, for every $j \in \mathscr{J}$ and every $b \in \mathscr{B}$, we have:

$$x_{jb} \in \{0,1\} \qquad \Rightarrow \qquad x_{jb} \leq 1$$

but they are implied by the constraints $\sum_{b \in \mathscr{B}} x_{jb} = 1$.

Optimal solutions of $\mathrm{LP}_{\mathrm{BPP}}$ and of its dual

Proposition 3

An optimal solution of the primal $\mathrm{LP}_{\mathrm{BPP}}$ *model is:*

$$\tilde{x}_{jb} = \frac{1}{m}, \qquad \forall j \in \mathscr{J},\ \forall b \in \mathscr{B},$$

$$\tilde{y}_b = \frac{\sum_{j \in \mathscr{J}} w_j}{m\, c}, \qquad \forall b \in \mathscr{B},$$

and an optimal solution of the dual $D(\mathrm{LP}_{\mathrm{BPP}})$ *model is:*

$$\tilde{\mu}_j = \frac{w_j}{c}, \quad \forall j \in \mathscr{J},$$

$$\tilde{\pi}_b = \frac{1}{c}, \quad \forall b \in \mathscr{B},$$

$$\tilde{\lambda}_b = 0, \quad \forall b \in \mathscr{B}.$$

The optimal value of $\mathrm{LP}_{\mathrm{BPP}}$

Corollary 2

The optimal value of the primal $\mathrm{LP}_{\mathrm{BPP}}$ *and dual* $D(\mathrm{LP}_{\mathrm{BPP}})$ *models is:*

$$z(\mathrm{LP}_{\mathrm{BPP}}) = \frac{\sum_{j \in \mathscr{J}} w_j}{c} = z(D(\mathrm{LP}_{\mathrm{BPP}}))$$

Proof

The proof is similar to the previous one. In addition, it is necessary to observe that

$$\tilde{y}_b = \frac{\sum_{j \in \mathscr{I}} w_j}{m\ c} \leq 1, \quad \forall b \in \mathscr{B},$$

since

$$\frac{\sum_j w_j}{c} \leq z(\mathrm{ILP_{BPP}}) \leq m,$$

where the last inequality holds because m is an upper bound on the number of bins in any optimal BPP solution.

Accordingly, the constraints $y_b \leq 1$ are also satisfied. □

Some important theoretical properties

- The following two propositions show that the ratio between $z(\mathrm{LP}_{\mathrm{BPP}})$ and $z(\mathrm{ILP}_{\mathrm{BPP}})$ is at least one half for every BPP instance.
- **By hypothesis** $z(\mathrm{ILP}_{\mathrm{BPP}}) > 1$, since not all items fit in a single bin.
- Furthermore, there exist instances for which the value one half can be asymptotically achieved as the capacity $c \to +\infty$.

The LP bound is at least one half

Proposition 4

For every BPP instance, we have:

$$\frac{z(\mathrm{LP_{BPP}})}{z(\mathrm{ILP_{BPP}})} \geq \frac{1}{2}$$

Proof (1/4)

- Consider an optimal solution $\tilde{\boldsymbol{x}}(\mathrm{ILP}_{\mathrm{BPP}})$ of value $z = z(\mathrm{ILP}_{\mathrm{BPP}})$ (number of used bins).
- We can assume, without loss of generality, that bins $1, 2, \ldots, z$ are the non-empty ones.
- In an optimal solution, no two bins can have total item weight at most $c/2$, since otherwise their contents could be merged into a single bin without exceeding the bin capacity c (contradicting optimality).

Proof (2/4)

If all non-empty bins have total item weight greater than $c/2$, then

$$\sum_{j \in \mathscr{I}} w_j = \sum_{b=1}^{z} \sum_{j \in \mathscr{I}} w_j \, \tilde{x}_{jb} > z \, \frac{c}{2}$$

$$\Longrightarrow \quad \frac{\frac{\sum_{j \in \mathscr{I}} w_j}{c}}{z} > \frac{1}{2} \quad \Longrightarrow \quad \frac{z(\mathrm{LP_{BPP}})}{z(\mathrm{ILP_{BPP}})} \geq \frac{1}{2}.$$

Proof (3/4)

Otherwise, suppose that there exists a bin with total item weight at most $c/2$. Renumbering bins if necessary, let it be bin z, so that

$$\sum_{j \in \mathscr{I}} w_j \, \tilde{x}_{jz} \leq \frac{c}{2}.$$

Since $z > 1$, there exists another bin say bin 1; by optimality, we must have

$$\sum_{j \in \mathscr{I}} w_j \, \tilde{x}_{j1} + \sum_{j \in \mathscr{I}} w_j \, \tilde{x}_{jz} \; > \; c,$$

and, by optimality, every bin in $\{2, 3, \ldots, z-1\}$ has total item weight $> c/2$.

Proof (4/4)

Hence,

$$\sum_{j\in\mathscr{I}} w_j = \sum_{b=1}^{z}\sum_{j\in\mathscr{I}} w_j\,\tilde{x}_{jb} = \underbrace{\sum_{b=2}^{z-1}\sum_{j\in\mathscr{I}} w_j\,\tilde{x}_{jb}}_{\geq \frac{c}{2}(z-2)} + \underbrace{\sum_{j\in\mathscr{I}} w_j\,\tilde{x}_{j1} + \sum_{j\in\mathscr{I}} w_j\,\tilde{x}_{jz}}_{>c}$$

$$> \frac{c}{2}(z-2) + c = z\,\frac{c}{2}$$

$$\Longrightarrow \quad \frac{\frac{\sum_{j\in\mathscr{I}} w_j}{c}}{z} > \frac{1}{2} \quad \Longrightarrow \quad \frac{z(\mathrm{LP_{BPP}})}{z(\mathrm{ILP_{BPP}})} \geq \frac{1}{2}.$$

□

The factor one half cannot be improved:
we now build instances whose ratio tends to $\frac{1}{2}$.

The factor one half is tight as $c \to +\infty$

Proposition 5

There are BPP instances such that:

$$\frac{z(\mathrm{LP_{BPP}})}{z(\mathrm{ILP_{BPP}})} \to \frac{1}{2} \qquad \textit{as} \qquad c \to +\infty$$

Proof (1/2)

Given $\vartheta \in \mathbb{Z}_{\geq 1}$, consider the BPP instance with bin capacity $c = 2\vartheta$ and n items having the following weights:

$$w_j = \vartheta + 1, \quad \forall j \in \mathscr{I}.$$

No bin can contain more than one item, thus each item must be packed alone in its own bin. Therefore, the BPP optimal value is

$$z(\mathrm{ILP_{BPP}}) = n.$$

For the LP relaxation we have:

$$z(\mathrm{LP_{BPP}}) = \frac{n(\vartheta + 1)}{2\,\vartheta}.$$

Proof (2/2)

It follows that:

$$\lim_{\vartheta \to +\infty} \frac{z(\mathrm{LP_{BPP}})}{z(\mathrm{ILP_{BPP}})} = \lim_{\vartheta \to +\infty} \frac{\frac{n(\vartheta+1)}{2\,\vartheta}}{n} = \lim_{\vartheta \to +\infty} \left(\frac{1}{2} + \underbrace{\frac{1}{2\,\vartheta}}_{\to 0} \right) = \frac{1}{2}.$$

Accordingly, we conclude that:

$$\frac{z(\mathrm{LP_{BPP}})}{z(\mathrm{ILP_{BPP}})} \to \frac{1}{2} \qquad \text{as} \qquad c \to +\infty.$$

□

The rounded LP bound

- The following two propositions show that the ratio between $\lceil z(\mathrm{LP}_{\mathrm{BPP}}) \rceil$ and $z(\mathrm{ILP}_{\mathrm{BPP}})$ is at least one half for every BPP instance.
- Furthermore, there exist instances for which the value one half can be asymptotically achieved as the number of items $n \to +\infty$.

The rounded LP bound is at least one half

Proposition 6

For every BPP instance, we have:

$$\frac{\lceil z(\mathrm{LP_{BPP}}) \rceil}{z(\mathrm{ILP_{BPP}})} \geq \frac{1}{2}$$

Proof (1/4)

- Consider an optimal solution $\tilde{\boldsymbol{x}}(\mathrm{ILP}_{\mathrm{BPP}})$ of value $z = z(\mathrm{ILP}_{\mathrm{BPP}})$ (number of used bins).
- We can assume, without loss of generality, that bins $1, 2, \ldots, z$ are the non-empty ones.
- In an optimal solution, no two bins can have total item weight at most $c/2$, since otherwise their contents could be merged into a single bin without exceeding the bin capacity c (contradicting optimality).

Proof (2/4)

If all non-empty bins have total item weight greater than $c/2$, then

$$\sum_{j \in \mathscr{I}} w_j = \sum_{b=1}^{z} \sum_{j \in \mathscr{I}} w_j \, \tilde{x}_{jb} > \frac{z\,c}{2} \quad \Longrightarrow \quad \frac{\sum_{j \in \mathscr{I}} w_j}{c} > \frac{z}{2}$$

$$\Longrightarrow \quad \left\lceil \frac{\sum_{j \in \mathscr{I}} w_j}{c} \right\rceil \geq \left\lceil \frac{z}{2} \right\rceil \geq \frac{z}{2}$$

$$\Longrightarrow \quad \frac{\left\lceil \frac{\sum_{j \in \mathscr{I}} w_j}{c} \right\rceil}{z} \geq \frac{1}{2} \quad \Longrightarrow \quad \frac{\lceil z(\mathrm{LP_{BPP}}) \rceil}{z(\mathrm{ILP_{BPP}})} \geq \frac{1}{2}.$$

Proof (3/4)

Otherwise, suppose that there exists a bin with total item weight at most $c/2$. Renumbering bins if necessary, let it be bin z. Then we have:

$$\sum_{j \in \mathscr{I}} w_j = \sum_{b=1}^{z} \sum_{j \in \mathscr{I}} w_j \tilde{x}_{jb} > \sum_{b=1}^{z-1} \sum_{j \in \mathscr{I}} w_j \tilde{x}_{jb} > (z-1)\frac{c}{2}$$

$$\Longrightarrow \quad z < 2 \sum_{j \in \mathscr{I}} \frac{w_j}{c} + 1.$$

Proof (4/4)

Since z must be an integer, this implies:

$$z \leq \left\lceil 2 \sum_{j \in \mathscr{I}} \frac{w_j}{c} \right\rceil \leq 2 \left\lceil \sum_{j \in \mathscr{I}} \frac{w_j}{c} \right\rceil .$$

Therefore:

$$\frac{\left\lceil \sum_{j \in \mathscr{I}} \frac{w_j}{c} \right\rceil}{z} \geq \frac{1}{2} \quad \Longrightarrow \quad \frac{\lceil z(\mathrm{LP_{BPP}}) \rceil}{z(\mathrm{ILP_{BPP}})} \geq \frac{1}{2}.$$

□

The factor one half is tight as $n \to +\infty$

Proposition 7

There are BPP instances such that:

$$\frac{\lceil z(\mathrm{LP_{BPP}}) \rceil}{z(\mathrm{ILP_{BPP}})} \to \frac{1}{2} \qquad \text{as} \qquad n \to +\infty$$

Proof (1/2)

Given $n \in \mathbb{Z}_{\geq 2}$ (take n even), consider the BPP instance with bin capacity $c = n$ and n items having the following weights:

$$w_j = \frac{c}{2} + 1 = \frac{n}{2} + 1, \qquad \forall j \in \mathscr{I}.$$

No bin can contain more than one item, since $w_j > c/2$ for every j. Thus each item must be packed alone in its own bin. Therefore, the BPP optimal value is

$$z(\mathrm{ILP}_{\mathrm{BPP}}) = n.$$

Proof (2/2)

For the LP relaxation we have:

$$z(\mathrm{LP_{BPP}}) = \frac{\sum_{j \in \mathscr{J}} w_j}{c} = \frac{n\left(\frac{n}{2}+1\right)}{n} = \frac{n}{2}+1,$$

hence $\lceil z(\mathrm{LP_{BPP}}) \rceil = \frac{n}{2}+1$ (since n is even). It follows that:

$$\lim_{n \to +\infty} \frac{\lceil z(\mathrm{LP_{BPP}}) \rceil}{z(\mathrm{ILP_{BPP}})} = \lim_{n \to +\infty} \frac{\frac{n}{2}+1}{n} = \lim_{n \to +\infty} \left(\frac{1}{2} + \underbrace{\frac{1}{n}}_{\to 0} \right) = \frac{1}{2}.$$

Accordingly, we conclude that:

$$\frac{\lceil z(\mathrm{LP_{BPP}}) \rceil}{z(\mathrm{ILP_{BPP}})} \to \frac{1}{2} \qquad \text{as} \qquad n \to +\infty.$$

□

Summary (1/2)

The $\mathrm{LP}_{\mathrm{BPP}}$ model:

- the binary domains are replaced by $x_{jb} \geq 0$, $y_b \leq 1$ and $y_b \geq 0$
- the constraints $x_{jb} \leq 1$ are valid but implied by $\sum_{b \in \mathscr{B}} x_{jb} = 1$
- on the example instance: $z(\mathrm{LP}_{\mathrm{BPP}}) = \frac{8}{5}$, the same lower bound as before

The optimal value:

- the same solutions as for $\mathrm{LP\text{-}0}_{\mathrm{BPP}}$, with $\tilde{\lambda}_b = 0$
- $z(\mathrm{LP}_{\mathrm{BPP}}) = \sum_j w_j / c$

Summary (2/2)

The quality of the bound:

- $z(\mathrm{LP}_{\mathrm{BPP}})/z(\mathrm{ILP}_{\mathrm{BPP}}) \geq \frac{1}{2}$ for every instance
- the value $\frac{1}{2}$ is attained asymptotically as $c \to +\infty$

The rounded bound:

- $\lceil z(\mathrm{LP}_{\mathrm{BPP}}) \rceil / z(\mathrm{ILP}_{\mathrm{BPP}}) \geq \frac{1}{2}$ for every instance
- the value $\frac{1}{2}$ is attained asymptotically as $n \to +\infty$

An ILP model for the Cutting Stock Problem

Introduction: An ILP model for the Cutting Stock Problem

- The problem of this section is the Cutting Stock Problem (CSP), defined in the chapter «Packing problems», with its reference instance
- Each item j must be packed in d_j identical copies
- We write the $\mathrm{ILP}_{\mathrm{CSP}}$ model: the same structure as the $\mathrm{ILP}_{\mathrm{BPP}}$ model, with integer variables that count the copies

The integer variables

- Given an upper bound $m \in \mathbb{Z}_{\geq 2}$ on the number of bins used in any optimal CSP solution, we introduce the following mn **integer variables**:

$$x_{jb} = \text{number of copies of item } j \text{ packed in bin } b$$

$$\forall j \in \mathscr{I}, \quad \forall b \in \underbrace{\{1, 2, \ldots, m\}}_{=\mathscr{B}},$$

The activation variables

- We also introduce the following m **binary variables**:

$$y_b = \begin{cases} 1 & \text{if bin } b \text{ is used} \\ 0 & \text{otherwise} \end{cases} \qquad \forall b \in \mathscr{B}.$$

- The total number $\sum_{j \in \mathscr{I}} d_j$ of copies is a valid value of m, and so is the number of bins of any feasible CSP solution.

The $\mathrm{ILP}_{\mathrm{CSP}}$ model

Using these variables, an Integer Linear Programming (ILP) model for the Cutting Stock Problem (CSP) reads as follows:

$$\min \quad \sum_{b \in \mathscr{B}} y_b$$

$$\text{subject to} \quad \sum_{b \in \mathscr{B}} x_{jb} = d_j, \qquad \forall j \in \mathscr{I},$$

$$-\sum_{j \in \mathscr{I}} w_j \, x_{jb} + c \, y_b \geq 0, \qquad \forall b \in \mathscr{B},$$

$$x_{jb} \in \mathbb{Z}_{\geq 0}, \qquad \forall j \in \mathscr{I}, \ \forall b \in \mathscr{B},$$

$$y_b \in \{0, 1\}, \qquad \forall b \in \mathscr{B}.$$

This ILP model is referred to as the $\mathrm{ILP}_{\mathrm{CSP}}$ model.

Description of the objective function and constraints

- The linear objective function minimizes the total number of used bins.
- The constraints $\sum_{b \in \mathscr{B}} x_{jb} = d_j$ ensure that exactly d_j copies of each item j are packed (n linear constraints).
- The capacity constraints link the integer variables with the activation variables and impose the capacity restrictions, as the capacity constraints of the $\mathrm{ILP}_{\mathrm{BPP}}$ model (m linear constraints).
- Finally, the constraints $x_{jb} \in \mathbb{Z}_{\geq 0}$ and $y_b \in \{0, 1\}$ define the variables of the ILP model.

The $\mathrm{ILP}_{\mathrm{CSP}}$ model and the $\mathrm{ILP}_{\mathrm{BPP}}$ model

- With $d_j = 1$ for every item, the constraints $\sum_{b \in \mathcal{B}} x_{jb} = d_j$ give $x_{jb} \in \{0, 1\}$, and the $\mathrm{ILP}_{\mathrm{CSP}}$ model is the $\mathrm{ILP}_{\mathrm{BPP}}$ model.
- The copies of an item are identical: a single integer variable counts them in each bin.
- The $\mathrm{ILP}_{\mathrm{BPP}}$ model on the extended set of items would use $m \sum_j d_j$ binary variables instead of mn integer ones.

Example (1/3): the $\mathrm{ILP}_{\mathrm{CSP}}$ model

- We consider bins of capacity $c = 10$ and a set $\mathscr{I} = \{1, 2, 3, 4\}$ of 4 items ($n = 4$), with the following weights and required copies:

	$j = 1$	$j = 2$	$j = 3$	$j = 4$
w_j	4	5	3	6
d_j	2	1	3	1

- This is the CSP instance of the chapter «Packing problems», where a solution with 3 bins is found by eye: hence $m = 3$ is a valid upper bound.

Example (2/3): the $\mathrm{ILP}_{\mathrm{CSP}}$ model

The $\mathrm{ILP}_{\mathrm{CSP}}$ model with $m = 3$ for this CSP instance is:

$$
\begin{aligned}
\text{min} \quad & y_1 + y_2 + y_3 \\
\text{subject to} \quad & x_{11} + x_{12} + x_{13} = 2 \\
& x_{21} + x_{22} + x_{23} = 1 \\
& x_{31} + x_{32} + x_{33} = 3 \\
& x_{41} + x_{42} + x_{43} = 1 \\
& -4\,x_{11} - 5\,x_{21} - 3\,x_{31} - 6\,x_{41} + 10\,y_1 \geq 0 \\
& -4\,x_{12} - 5\,x_{22} - 3\,x_{32} - 6\,x_{42} + 10\,y_2 \geq 0 \\
& -4\,x_{13} - 5\,x_{23} - 3\,x_{33} - 6\,x_{43} + 10\,y_3 \geq 0 \\
& x_{11},\ x_{12},\ x_{13},\ \ldots,\ x_{43} \in \mathbb{Z}_{\geq 0} \\
& y_1,\ y_2,\ y_3 \in \{0, 1\}
\end{aligned}
$$

Example (3/3): the $\mathrm{ILP}_{\mathrm{CSP}}$ model

An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{ILP}_{\mathrm{CSP}})$ is:

$$\tilde{x}_{11} = \tilde{x}_{41} = 1, \quad \tilde{x}_{32} = 3, \quad \tilde{x}_{13} = \tilde{x}_{23} = 1, \quad \tilde{y}_1 = \tilde{y}_2 = \tilde{y}_3 = 1,$$

with all the other variables equal to 0.

- The optimal value is $z(\mathrm{ILP}_{\mathrm{CSP}}) = 3$, as found in the chapter «Packing problems».
- The bins contain one copy of items 1 and 4 (weight 10), three copies of item 3 (weight 9), and one copy of items 1 and 2 (weight 9).
- The optimal solution is not unique: the solution of the chapter «Packing problems», with the bins $\{4, 1\}$, $\{1, 3, 3\}$ and $\{2, 3\}$, also uses 3 bins.

Summary

The $\mathrm{ILP}_{\mathrm{CSP}}$ model:

- mn integer variables x_{jb}, the copies of item j in bin b, and m binary variables y_b
- the demands d_j on the right-hand side of the constraints $\sum_b x_{jb} = d_j$, the same capacity constraints as the $\mathrm{ILP}_{\mathrm{BPP}}$ model
- with $d_j = 1$ for every item it is the $\mathrm{ILP}_{\mathrm{BPP}}$ model

The reference CSP instance:

- $z(\mathrm{ILP}_{\mathrm{CSP}}) = 3$, with more than one optimal solution

Typical mistakes

Introduction: Typical mistakes

- We collect the most frequent mistakes on the packing models
- They concern the choice of m, the use of the LP bound, the meaning of the variables and the symmetry of the solutions
- Recognizing them helps to write correct models and correct bounds

Typical mistakes (1/3)

Observation 1: Typical mistakes

- ***Choosing*** *m* ***too small.*** *The number m of bins must be an upper bound on the number of bins of an optimal solution: with* $m < z(\mathrm{ILP}_{\mathrm{BPP}})$ *the model has no feasible solution. The value* $m = n$ *is always valid.*
- ***Forgetting to round up.*** *The optimal value of the* $\mathrm{ILP}_{\mathrm{BPP}}$ *model is an integer, so the lower bound from the relaxations is* $\lceil \sum_j w_j / c \rceil$*; in the example instance it is* $\lceil \frac{8}{5} \rceil = 2$*, not* $\frac{8}{5}$*.*

Typical mistakes (2/3)

Observation: Typical mistakes

- ***Believing that the model counts only non-empty bins.*** *The constraints allow $y_b = 1$ for an empty bin: it is the objective function that switches such bins off, in every optimal solution.*
- ***Ignoring the symmetry.*** *Permuting the indices of the bins gives another optimal solution with the same packing: the $\mathrm{ILP}_{\mathrm{BPP}}$ model has many equivalent optimal solutions.*

Typical mistakes (3/3)

Observation: Typical mistakes

- ***Expecting a better relaxation from*** $y_b \leq 1$. *The second relaxation has the same optimal value as the first one: the upper bounds* $y_b \leq 1$ *do not cut off the optimal solution of the first relaxation.*
- ***Declaring binary the variables of the cutting stock model.*** *In the* $\mathrm{ILP}_{\mathrm{CSP}}$ *model* x_{jb} *counts the copies of item* j *in bin* b*: with* $x_{jb} \in \{0, 1\}$ *the three copies of item 3 of the example could not share a bin.*

Summary

The model:

- m must be an upper bound on the optimal number of bins: $m = n$ is always valid
- empty bins are switched off by the objective function, not by the constraints
- the indices of the bins can be permuted: many equivalent optimal solutions

The bounds and the CSP:

- round the LP bound up; $y_b \leq 1$ does not improve it
- in the $\mathrm{ILP}_{\mathrm{CSP}}$ model the variables x_{jb} are integer, not binary

In summary

Introduction: In summary

- We collect the key ideas of the chapter
- The two ILP models, the two LP relaxations and the quality of their bound
- A dashboard with the bounds computed on the example instance

Summary (1/3)

- The $\mathrm{ILP}_{\mathrm{BPP}}$ model has $m \cdot n$ assignment variables x_{jb} and m activation variables y_b; the capacity constraints link them.
- Both LP relaxations have optimal value $z(\mathrm{LP}_{\mathrm{BPP}}) = \sum_j w_j/c$, with dual solution $\tilde{\mu}_j = w_j/c$ and $\tilde{\pi}_b = 1/c$.
- Since the optimal value is an integer, $\lceil z(\mathrm{LP}_{\mathrm{BPP}}) \rceil$ is a lower bound on the number of bins.

Summary (2/3)

- The ratio between $z(\mathrm{LP}_{\mathrm{BPP}})$ and $z(\mathrm{ILP}_{\mathrm{BPP}})$ is at least $\frac{1}{2}$, and the value $\frac{1}{2}$ is asymptotically attained; the same holds for $\lceil z(\mathrm{LP}_{\mathrm{BPP}}) \rceil$.

- The $\mathrm{ILP}_{\mathrm{CSP}}$ model has the structure of the $\mathrm{ILP}_{\mathrm{BPP}}$ model, with mn integer variables x_{jb} that count the copies and the demands d_j on the right-hand side; on the CSP instance of the chapter «Packing problems» its optimal value is 3.

Summary (3/3)

Dashboard of the example instance. Bounds on $z(\mathrm{ILP}_{\mathrm{BPP}})$ computed in this chapter:

Bound	Value	From
First relaxation	$\frac{8}{5}$	$z(\mathrm{LP\text{-}0}_{\mathrm{BPP}})$
Second relaxation	$\frac{8}{5}$	$z(\mathrm{LP}_{\mathrm{BPP}})$
Lower bound	2	$\lceil \frac{8}{5} \rceil$
Optimal value	2	patterns $\{1, 3\}$ and $\{2\}$

Here the rounded bound is already equal to the optimal value.